



Computer Systems Security

Dr. Sandra Wahid

Multiple Encryption and Triple DES

- Since DES is vulnerable to brute-force attacks and once it was the most widely used symmetric cipher, it has been largely replaced by stronger encryption schemes.
- Two approaches have been taken:
 1. Design a **new algorithm** such as AES
 2. Preserve the existing investment in software and equipment, using **multiple encryption** with DES and multiple keys → **Triple-DES** is the widely chosen form.

Double DES

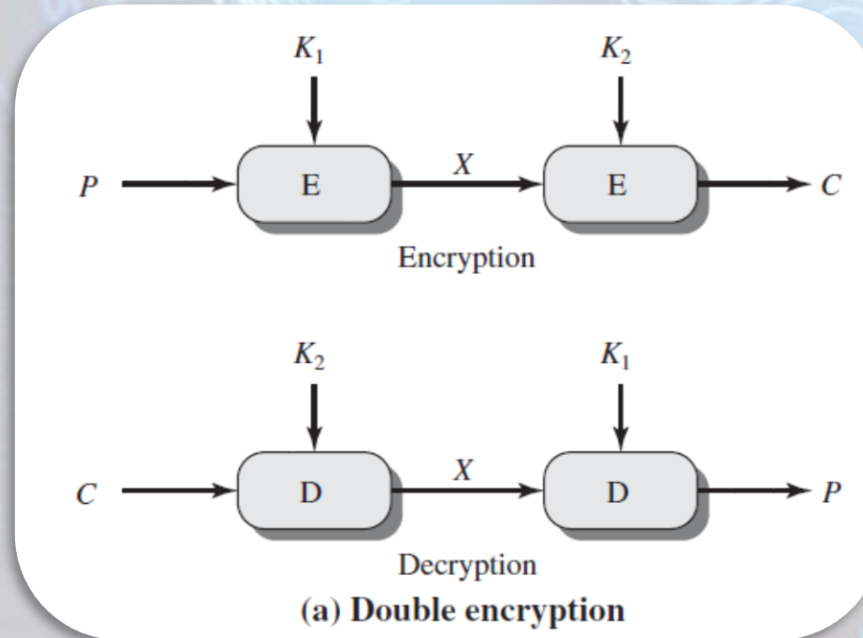
- Two encryption stages and two keys:

$$C = E(K_2, E(K_1, P))$$

- Decryption requires that the keys be applied in **reverse order**:

$$P = D(K_1, D(K_2, C))$$

- This scheme apparently involves a **key length of $56 \times 2 = 112$ bits**, and should result in a dramatic increase in cryptographic strength.



Double DES

Meet-in-the-middle attack:

- It is one way to attack double DES that does not depend on any particular property of DES but that will work against any block encryption cipher.

How it works?

- We have $C = E(K_2, E(K_1, P))$ so $X = E(K_1, P) = D(K_2, C)$
- Given a known pair: (P, C) , first, encrypt P for all 2^{56} possible values of K_1 . Store these results in a table and then sort the table by the values of X .
- Next, decrypt C using all 2^{56} possible values of K_2 .
- As each decryption is produced, check the result against the table for a match.
 - If a match occurs, then test the two resulting keys against a new known plaintext-ciphertext pair.
 - If the two keys produce the correct ciphertext, accept them as the correct keys.

Triple DES

- An obvious counter to the meet-in-the-middle attack is to use three stages of encryption with different keys.

Triple DES with Two Keys

Triple DES with Three Keys

1. Triple DES with Two Keys:

$$C = E(K_1, D(K_2, E(K_1, P)))$$
$$P = D(K_1, E(K_2, D(K_1, C)))$$

- There is no cryptographic significance to the use of decryption for the second stage.
- Its only advantage is that it allows users of 3DES to decrypt data encrypted by users of the older single DES:

$$C = E(K_1, D(K_1, E(K_1, P))) = E(K_1, P)$$
$$P = D(K_1, E(K_1, D(K_1, C))) = D(K_1, C)$$

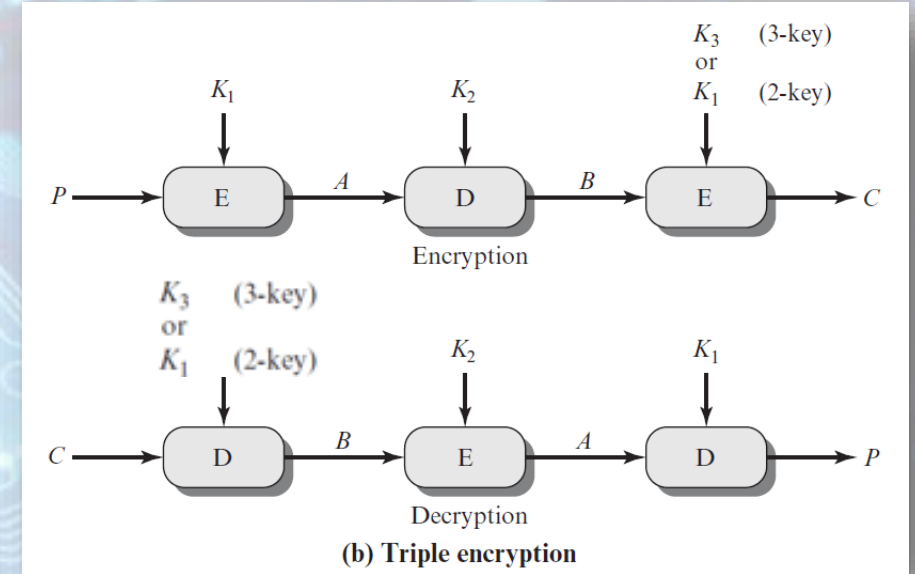
Backward compatibility with DES is provided by putting $K_1 = K_2$

2. Triple DES with Three Keys:

$$C = E(K_3, D(K_2, E(K_1, P)))$$

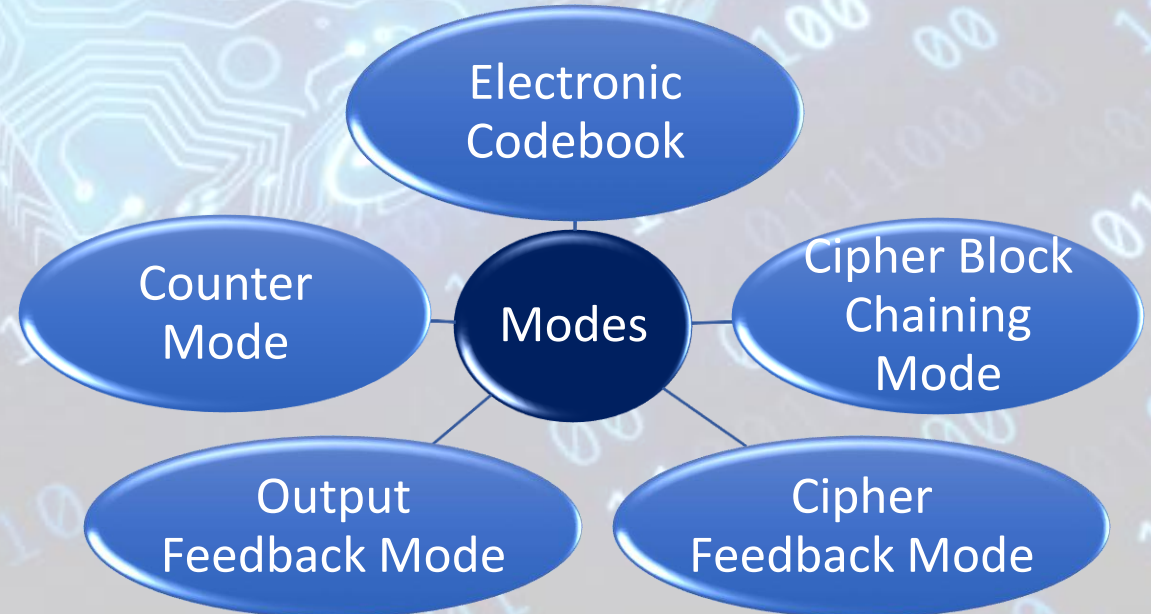
$$P = D(K_1, E(K_2, D(K_3, C)))$$

Backward compatibility with DES is provided by putting $K_1 = K_2$ or $K_3 = K_2$



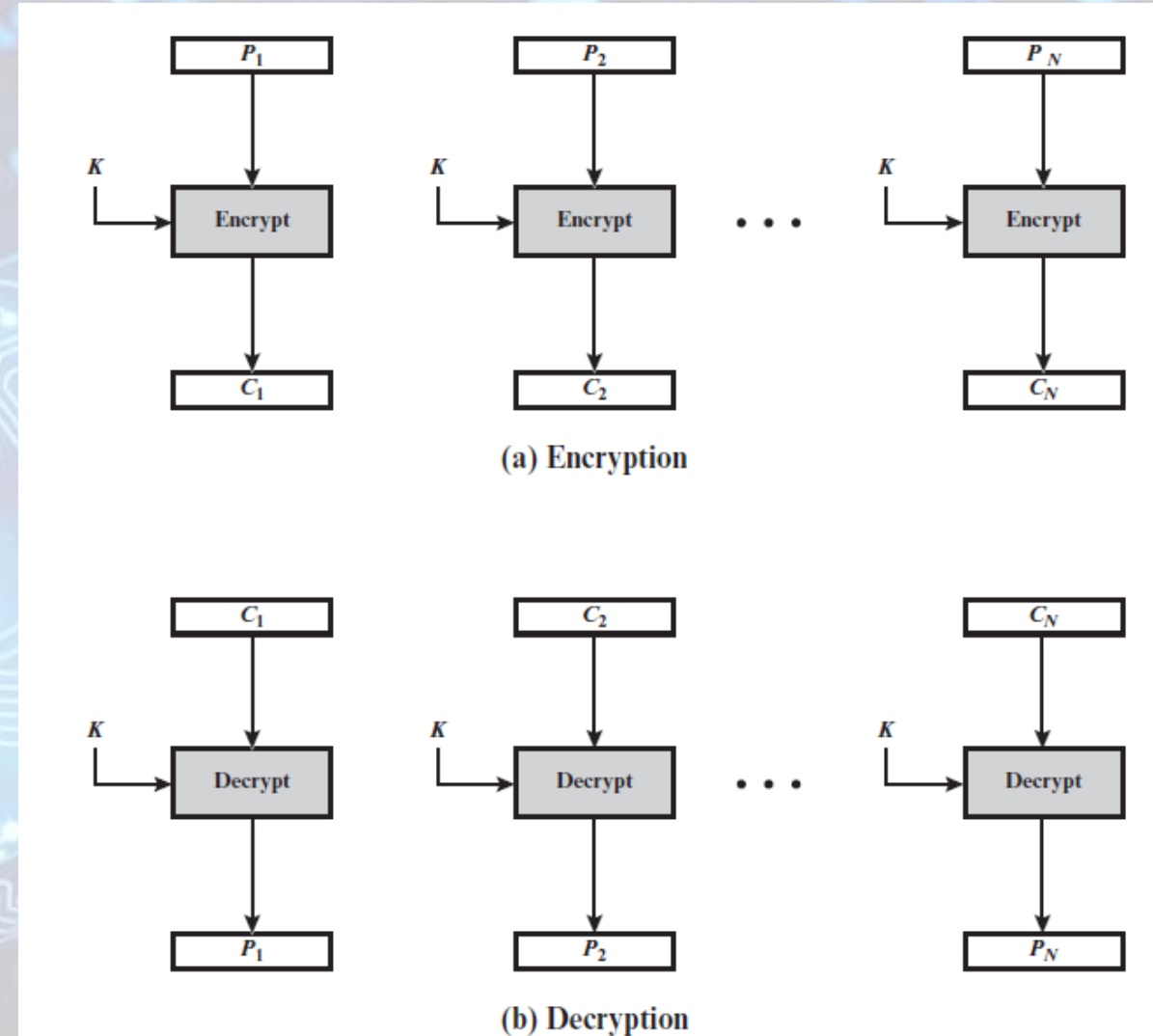
Block Cipher Modes of Operation

- A block cipher takes a fixed-length block of text of length b bits and a key as input and produces a b -bit block of ciphertext.
- If the amount of plaintext to be encrypted is greater than b bits, then the block cipher can still be used by **breaking the plaintext up** into b -bit blocks.
- When multiple blocks of plaintext are encrypted using the same key, a number of security issues arise.
- To apply a block cipher in a variety of applications, **five modes** of operation have been defined:
 - Have block and stream modes.
 - Can be used with any block cipher including triple DES and AES.



Electronic Codebook (ECB)

- Simplest mode.
- Plaintext is handled **one block at a time** and each block of plaintext is encrypted using the **same key**.
- The term codebook is used because, for a given key, there is a unique ciphertext for every b-bit block of plaintext. Therefore, we can imagine a gigantic codebook in which there is an entry for every possible b-bit plaintext pattern showing its corresponding ciphertext.
- If the last plaintext block is less than b-bits then padding is used.
- Decryption is performed one block at a time, always using the same key.



ECB	$C_j = E(K, P_j) \quad j = 1, \dots, N$	$P_j = D(K, C_j) \quad j = 1, \dots, N$
-----	---	---

ECB Analysis

- The most significant characteristic of ECB is that if the same b-bit block of plaintext appears more than once in the message, it always produces the **same** ciphertext.
- For lengthy messages, the ECB mode may not be secure → If the message is highly structured, it may be possible for a cryptanalyst to exploit these **regularities**.
- Can be used for secure transmission of single values (e.g., an encryption key)

Evaluation Criteria for Block Cipher Modes of Operation

- **Overhead**: The additional operations for the encryption and decryption operation when compared to encrypting and decrypting in the ECB mode.
- **Error recovery**: The property that an error in the i th ciphertext block is inherited by only a few plaintext blocks after which the mode resynchronizes.
- **Error propagation**: The property that an error in the i th ciphertext block is inherited by the i th and all subsequent plaintext blocks. What is meant here is a bit error that occurs in the transmission of a ciphertext block, not a computational error in the encryption of a plaintext block.
- **Diffusion**: How the plaintext statistics are reflected in the ciphertext.
- **Security**: Whether or not the ciphertext blocks leak information about the plaintext blocks.

Cipher Block Chaining (CBC)

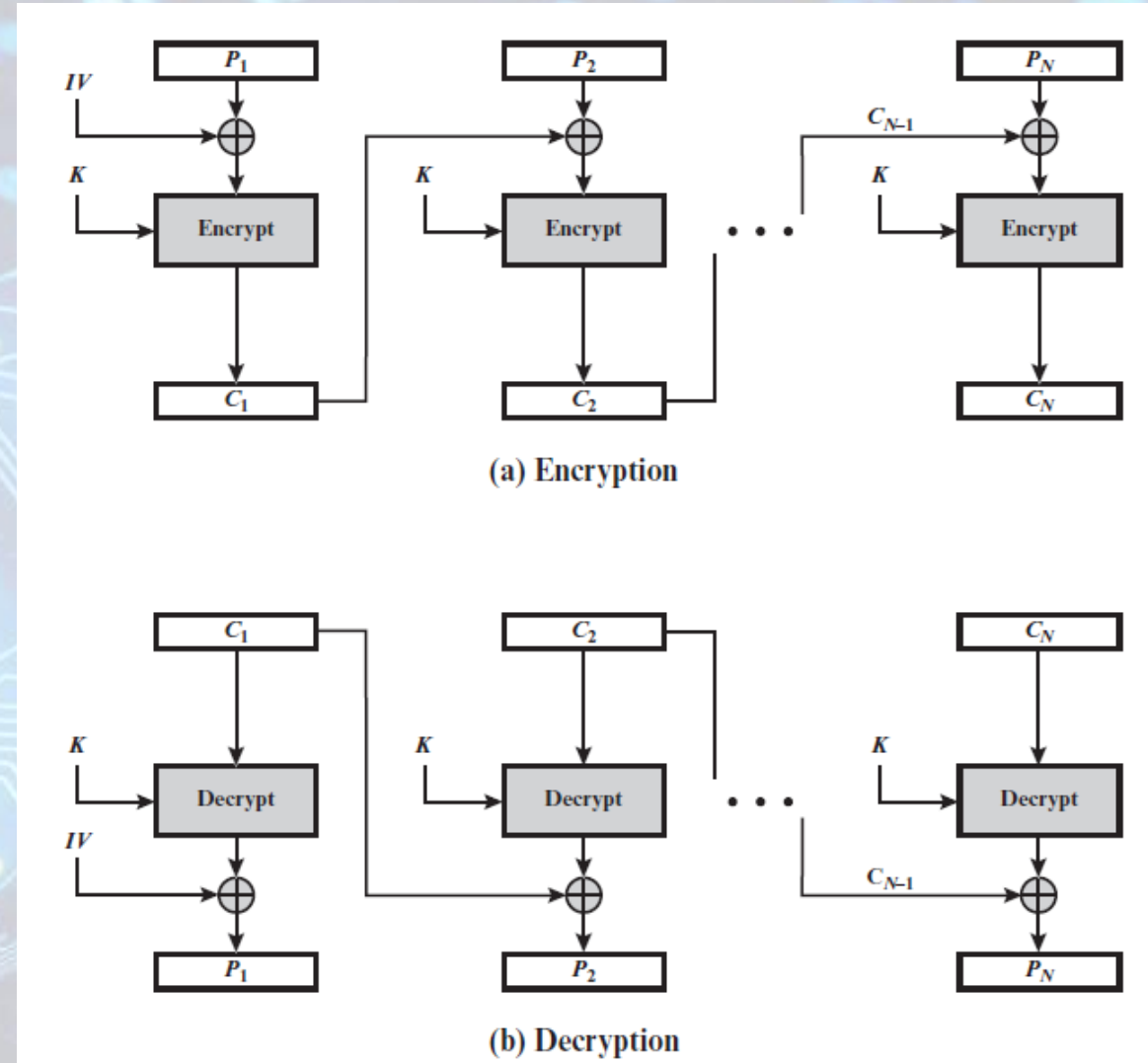
- To overcome the security deficiencies of ECB, we would like a technique in which the same plaintext block, if repeated, produces different ciphertext blocks.
- The input to the encryption algorithm is the XOR of the current plaintext block and the preceding ciphertext block.
- The same key is used for each block.
- Requires that the last block be padded to a full b bits if it is a partial block.
- Why it works?

$$C_j = E(K, [C_{j-1} \oplus P_j])$$

$$D(K, C_j) = D(K, E(K, [C_{j-1} \oplus P_j]))$$

$$D(K, C_j) = C_{j-1} \oplus P_j$$

$$C_{j-1} \oplus D(K, C_j) = C_{j-1} \oplus C_{j-1} \oplus P_j = P_j$$



CBC	$C_1 = E(K, [P_1 \oplus IV])$	$P_1 = D(K, C_1) \oplus IV$
	$C_j = E(K, [P_j \oplus C_{j-1}]) \quad j = 2, \dots, N$	$P_j = D(K, C_j) \oplus C_{j-1} \quad j = 2, \dots, N$

CBC Analysis

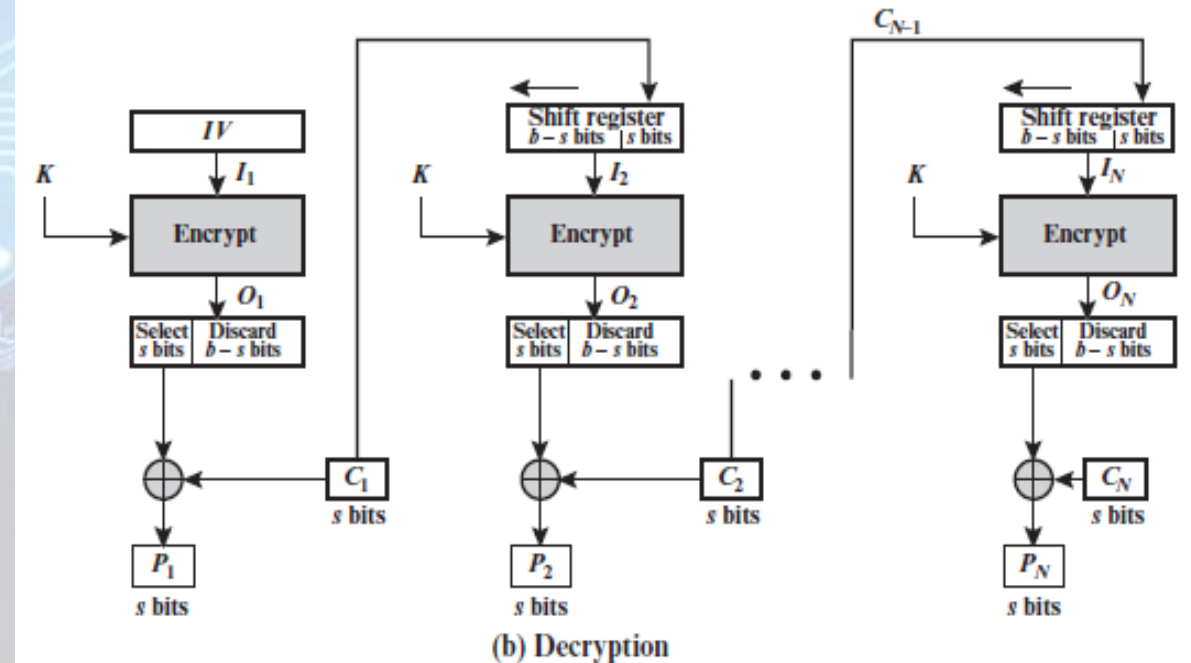
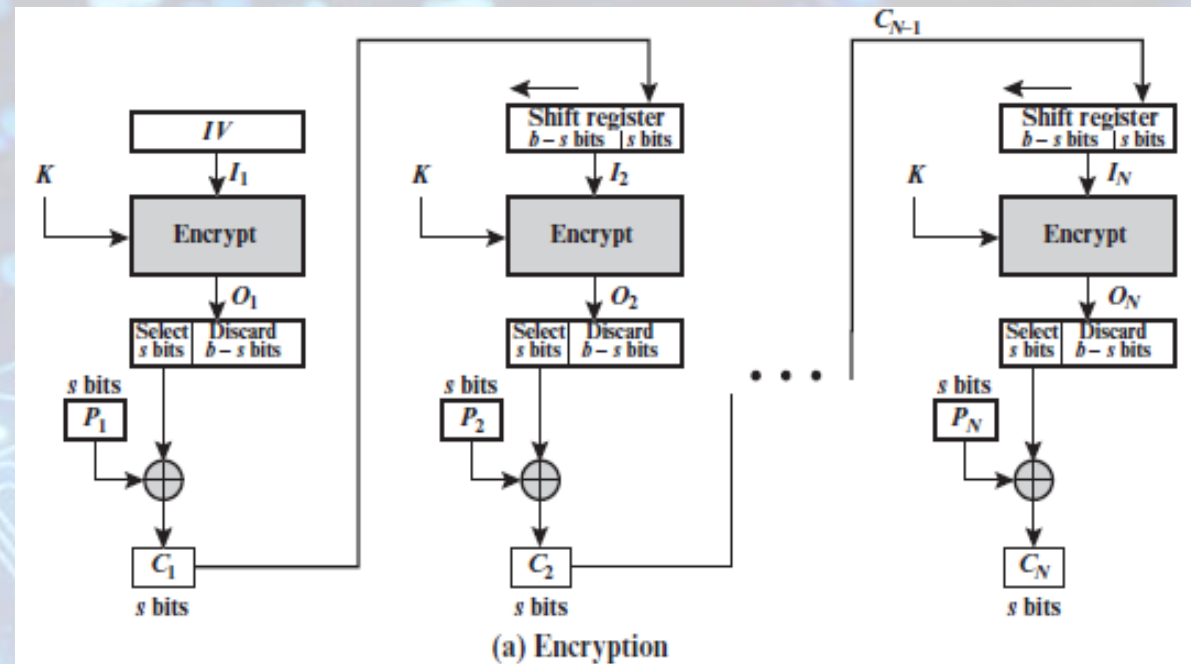
- The **Initialization Vector (IV)** must be known to both the sender and receiver but be unpredictable by a third party.
- The IV should be protected against unauthorized changes → This could be done by sending the IV using ECB encryption.
 - If an opponent is able to fool the receiver into using a different value for IV, then the opponent is able to invert selected bits in the first block of plaintext.
- CBC is an appropriate mode for encrypting messages of length greater than b bits because of its chaining mechanism.
- In addition to its use to achieve confidentiality, the CBC mode can be used for authentication.

Stream Modes of Operation

- For any block cipher, encryption is performed on a block of b bits.
- However, it is possible to **convert a block cipher into a stream cipher**, using one of the three modes
 - cipher feedback (CFB) mode, output feedback (OFB) mode, and counter (CTR) mode.
- A stream cipher:
 - eliminates the need to pad a message to be an integral number of blocks.
 - can operate in real time.
 - if a character stream is being transmitted, each character can be encrypted and transmitted immediately using a character-oriented stream cipher.

Cipher Feedback (CFB)

- It is assumed that the unit of transmission is s bits; a common value is $s = 8$.
- Rather than blocks of b bits, the plaintext is divided into segments of s bits.
- The leftmost (most significant) s bits of the output of the encryption function are XORed with the s segment plaintext to produce the s -bits ciphertext, which is then transmitted.
- The contents of the shift register are shifted left by s bits, and C is placed in the rightmost (least significant) s bits of the shift register.
- For decryption, the same scheme is used, except that the received C unit is XORed with the output of the encryption function to produce the P unit.



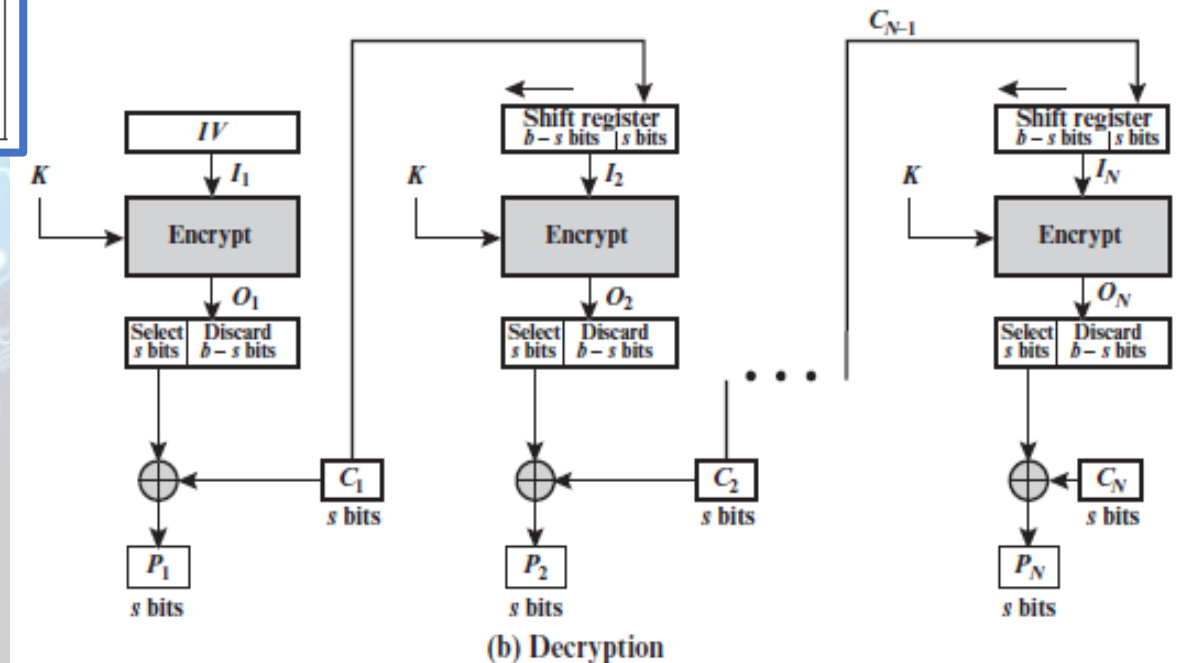
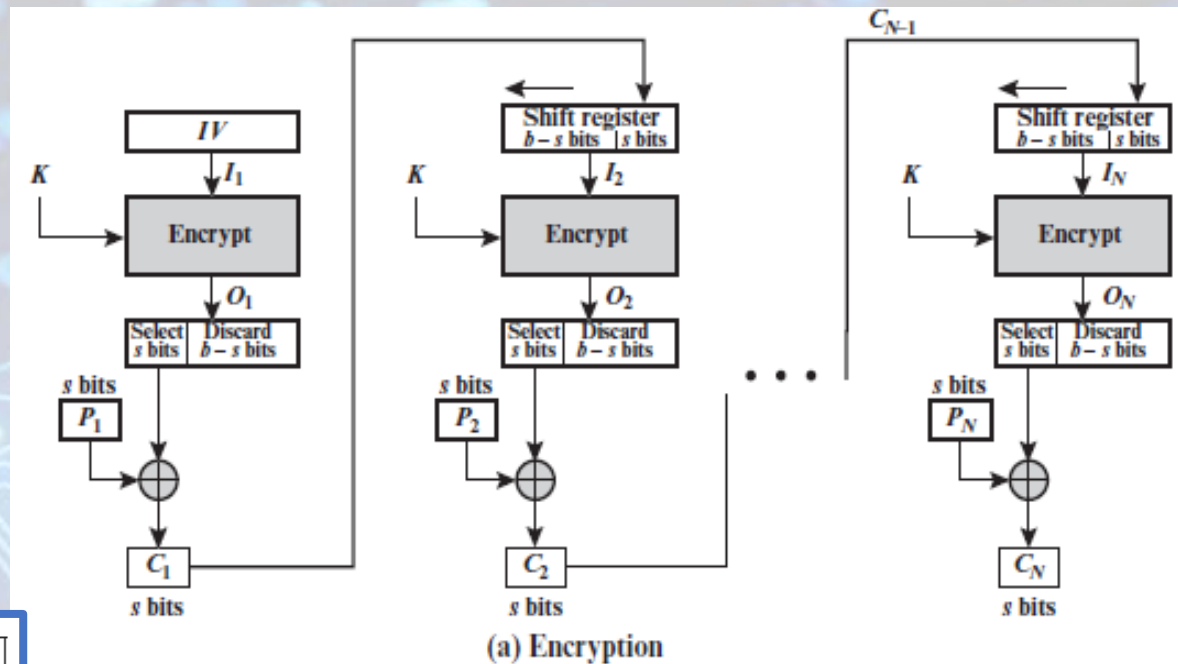
Cipher Feedback (CFB)

- Note that the encryption function is used not the decryption function. Why??

$$C_1 = P_1 \oplus \text{MSB}_s[E(K, IV)]$$

Therefore, by rearranging terms:

$$P_1 = C_1 \oplus \text{MSB}_s[E(K, IV)]$$



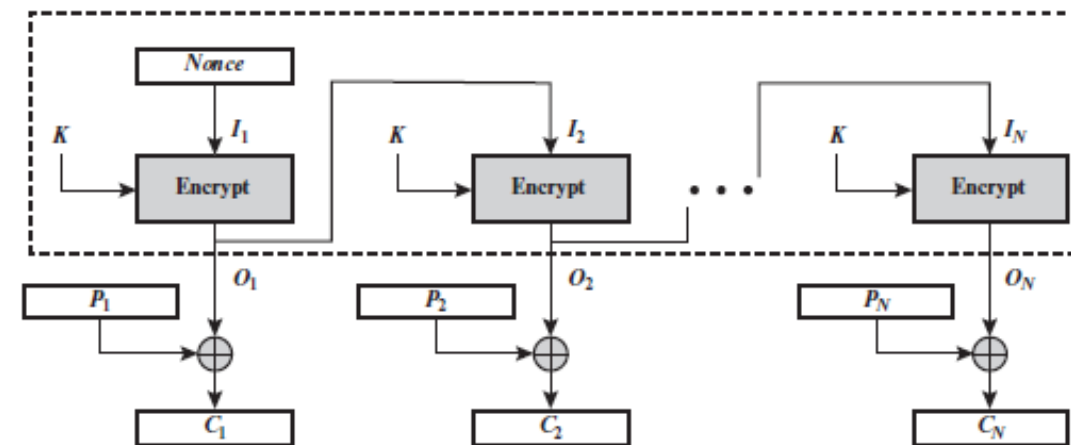
CFB	$I_1 = IV$	$I_1 = IV$
	$I_j = \text{LSB}_{b-s}(I_{j-1}) \parallel C_{j-1} \quad j = 2, \dots, N$	$I_j = \text{LSB}_{b-s}(I_{j-1}) \parallel C_{j-1} \quad j = 2, \dots, N$
	$O_j = E(K, I_j) \quad j = 1, \dots, N$	$O_j = E(K, I_j) \quad j = 1, \dots, N$
	$C_j = P_j \oplus \text{MSB}_s(O_j) \quad j = 1, \dots, N$	$P_j = C_j \oplus \text{MSB}_s(O_j) \quad j = 1, \dots, N$

CFB Analysis

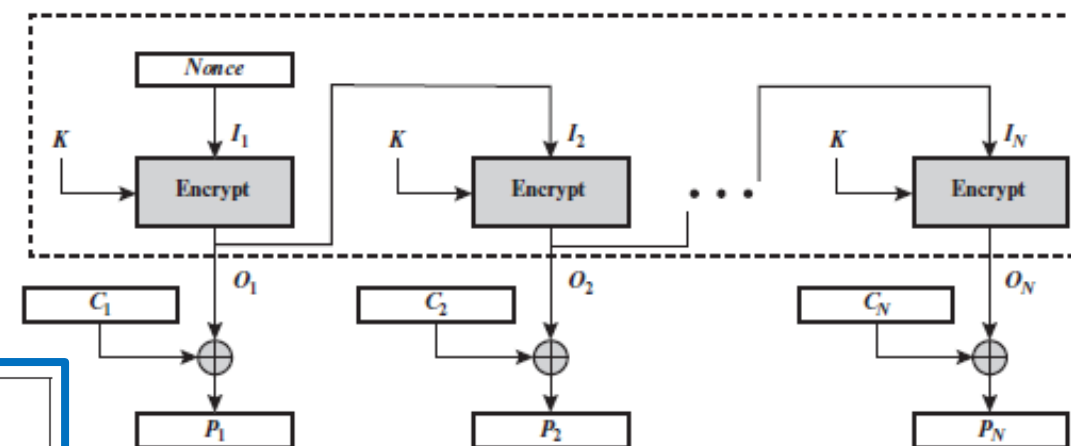
- In CFB encryption, like CBC encryption, the input block to each forward cipher function (except the first) depends on the result of the previous forward cipher function; therefore, multiple forward cipher operations **cannot** be performed in **parallel**.
- In CFB decryption, the required forward cipher operations can be performed in **parallel** since all the C's are sent to the receiver.

Output Feedback (OFB)

- Similar in structure to CFB.
 - For OFB, the output of the encryption function is fed back to become the input for encrypting the next block of plaintext.
 - In CFB, the output of the XOR unit is fed back to become input for encrypting the next block.
 - The other difference is that the OFB mode operates on full blocks of plaintext and ciphertext, whereas CFB operates on an s-bit subset.
- If the last block of plaintext contains u bits ($u < b$), the most significant u bits of the last output block O_N are used for the XOR operation and the remaining $b - u$ bits of O_N are discarded $\rightarrow$ no need for padding.



(a) Encryption

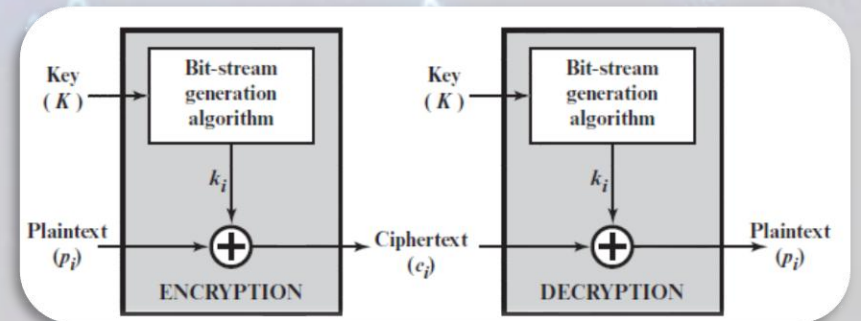


(b) Decryption

OFB	$I_1 = \text{Nonce}$	$I_1 = \text{Nonce}$
	$I_j = O_{j-1} \quad j = 2, \dots, N$	$I_j = O_{j-1} \quad j = 2, \dots, N$
	$O_j = E(K, I_j) \quad j = 1, \dots, N$	$O_j = E(K, I_j) \quad j = 1, \dots, N$
	$C_j = P_j \oplus O_j \quad j = 1, \dots, N - 1$	$P_j = C_j \oplus O_j \quad j = 1, \dots, N - 1$
	$C_N^* = P_N^* \oplus \text{MSB}_u(O_N)$	$P_N^* = C_N^* \oplus \text{MSB}_u(O_N)$

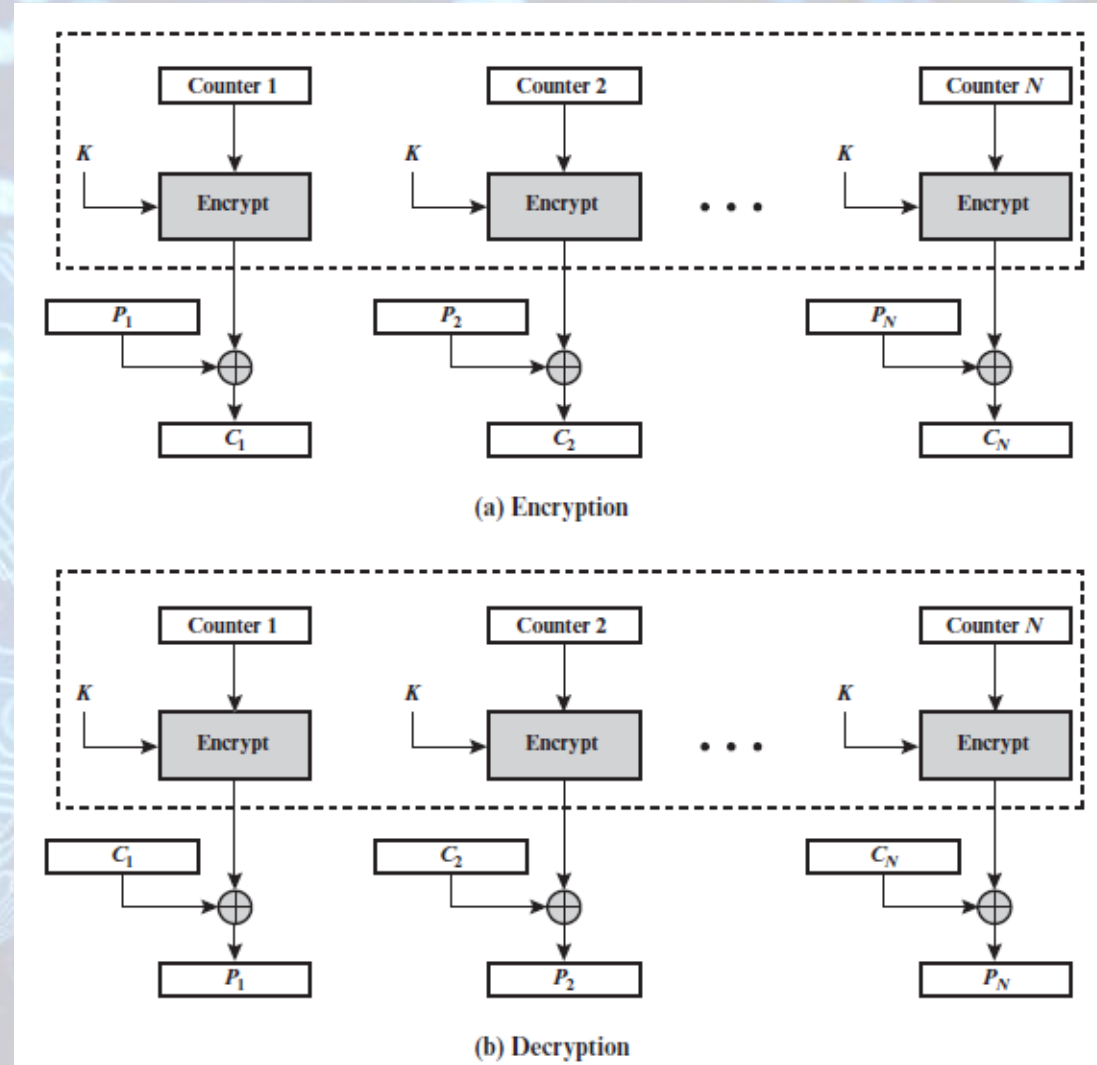
OFB Analysis

- As with CBC and CFB, the OFB mode requires an initialization vector.
 - In the case of OFB, the IV must be a **nonce**: the IV must be unique to each execution of the encryption operation. Why??
 - Encryption output blocks, O_i , depends only on the key and the IV and does not depend on the plaintext. Therefore, for a given key and IV, the stream of output bits used to XOR with the stream of plaintext bits is fixed.
- One advantage of the OFB method is that **bit errors in transmission do not propagate**. For example, if a bit error occurs in C_1 , only the recovered value of P_1 is affected; subsequent plaintext units are not corrupted. With CFB, C_1 also serves as input to the shift register and therefore causes additional corruption downstream.
- The disadvantage of OFB is that it is more **vulnerable to a message stream modification attack** than is CFB.
 - Complementing a bit in the ciphertext complements the corresponding bit in the recovered plaintext. Thus, controlled changes to the recovered plaintext can be made.
- OFB **has the structure of a typical stream cipher**, because the cipher generates a stream of bits as a function of an initial value and a key (independent of the plaintext), and that stream of bits is XORed with the plaintext bits.



Counter (CTR)

- A counter equal to the plaintext block size is used.
- Typically, the counter is initialized to some value and then incremented by 1 for each subsequent block.
- For encryption, the counter is encrypted and then XORed with the plaintext block to produce the ciphertext block → there is no chaining.
- For decryption, the same sequence of counter values is used. Thus, the initial counter value must be made available for decryption.
- No need for padding.



CTR

$$C_j = P_j \oplus E(K, T_j) \quad j = 1, \dots, N - 1$$

$$C_N^* = P_N^* \oplus \text{MSB}_u[E(K, T_N)]$$

$$P_j = C_j \oplus E(K, T_j) \quad j = 1, \dots, N - 1$$

$$P_N^* = C_N^* \oplus \text{MSB}_u[E(K, T_N)]$$

CTR Analysis

- As with the OFB mode, the initial counter value must be a **nonce**.
 - T1 must be different for all of the messages encrypted using the same key.
- Further, all T_i values across all messages must be unique.
 - If, contrary to this requirement, a counter value is used multiple times, then the confidentiality of all of the plaintext blocks corresponding to that counter value may be compromised.
 - In particular, if any plaintext block that is encrypted using a given counter value is known, then the output of the encryption function can be determined easily from the associated ciphertext block ($O = P \text{ XOR } C$). This output allows any other plaintext blocks that are encrypted using the same counter value to be easily recovered from their associated ciphertext blocks ($P' = O \text{ XOR } C'$).
- One way to ensure the uniqueness of counter values is to continue to **increment the counter value by 1 across messages**.
 - The first counter value of each message is one more than the last counter value of the preceding message.
- Same as OFB, CTR **has the structure of a typical stream cipher**, because the cipher generates a stream of bits as a function of a counter and a key (independent of the plaintext), and that stream of bits is XORed with the plaintext bits.

CTR Analysis

- Advantages:

- **Simplicity**

- Encryption/decryption can be done in **parallel** unlike chaining modes where the algorithm must complete the computation on one block before beginning on the next block.

- **Preprocessing** can be done: the execution of the underlying encryption algorithm does not depend on the input plaintext or ciphertext so the output of the encryption boxes can be prepared until the input P or C arrives, then the only remaining computation will be a series of XORs.

- **Random access** can be achieved: a certain plaintext or ciphertext block is processed without the need to process any preceding blocks.

- **Provable security**: CTR is at least as secure as the other modes.

→ interest in the counter (CTR) mode has increased recently.

Summary of Block Cipher Modes of Operation

Mode	Description	Typical Application
Electronic Codebook (ECB)	Each block of plaintext bits is encoded independently using the same key.	• Secure transmission of single values (e.g., an encryption key)
Cipher Block Chaining (CBC)	The input to the encryption algorithm is the XOR of the next block of plaintext and the preceding block of ciphertext.	• General-purpose block-oriented transmission • Authentication
Cipher Feedback (CFB)	Input is processed s bits at a time. Preceding ciphertext is used as input to the encryption algorithm to produce pseudorandom output, which is XORed with plaintext to produce next unit of ciphertext.	• General-purpose stream-oriented transmission • Authentication
Output Feedback (OFB)	Similar to CFB, except that the input to the encryption algorithm is the preceding encryption output, and full blocks are used.	• Stream-oriented transmission over noisy channel (e.g., satellite communication)
Counter (CTR)	Each block of plaintext is XORed with an encrypted counter. The counter is incremented for each subsequent block.	• General-purpose block-oriented transmission • Useful for high-speed requirements



Thank You